Samuel Appiah Kubi

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SUMMARY

Aspiring researcher in geospatial artificial intelligence with a strong foundation in geomatic engineering, specializing in the application of machine learning to satellite remote sensing data. Experienced in building end-to-end analytical systems that integrate computer vision, spatial analytics, and cloud platforms for agricultural and infrastructure challenges in West Africa. Proficient in developing scalable data pipelines with Python and PostGIS, with a proven ability to identify and address algorithmic biases, including shortcut learning, when models fail to generalize across diverse environments. Committed to advancing robust, equitable, and physically-informed geospatial foundation models for real-world deployment.

RESEARCH INTEREST

- Developing robust machine learning models that generalize across diverse landscapes for land use/cover mapping
- Integrating satellite remote sensing with cloud computing for large-scale agricultural and urban monitoring
- Understanding and mitigating algorithmic bias when models trained in one region fail in another
- Bridging remote sensing data with physical reality through sensor calibration and multi-source data fusion
- Building scalable geospatial data pipelines that support irrigation, urbanization, and land use change analysis

EDUCATION AND CERTIFICATIONS

Kwame Nkrumah University of Science and Technology (KNUST) Jan 2020 – Aug 2024
Bachelor of Science (BS) in Geomatic Engineering

- Graduated with **First Class Honours** (CGPA of 71.74%)
- **Major Courses:** Remote Sensing, GIS, Geodesy, Photogrammetry, Programming, Astronomy, Cadastral Survey, Engineering Survey, and Cartography

LANGUAGES

Languages

- **English:** Fluent (Professional Proficiency)

RESEARCH EXPERIENCE

Undergraduate Thesis: Phone Camera Calibration for Mapping Applications, a key component of my research on Smartphone-Based Road Condition Monitoring
Technologies: Python, OpenCV, YOLOv8, IMU, Sensor Fusion, GPS
Supervisor: Prof. Samuel Ato Andam-Akorful

- Developed an integrated system combining camera calibration, computer vision, and IMU sensors to estimate physical pothole dimensions from mobile platforms.
- Observed significant model degradation when moving from highways to unpaved rural roads, identifying model generalization challenges across diverse environments.
- Noted vision model relied on visual shortcuts (paved texture, lane markings) while IMU data provided physically-grounded measurements consistent across environments.
- Created multi-modal datasets logging pothole dimensions, IMU readings, and GPS coordinates for geospatial analysis.
- Recognized that integrating physical sensors with vision offers a pathway to more robust predictions, directly informing interest in physically informed geospatial machine learning.

CONFERENCE AND PRESENTATION

Geospatial Summer Camp — Town Mapping Project
Kwame Nkrumah University of Science and Technology (KNUST)
Lead: Prof. Samuel Ato Andam-Akorful

- Participated in an intensive two-week field camp focused on advanced geospatial data acquisition and town planning applications in Ayigya township.
- Conducted hands-on data collection of natural and man-made features using state-of-the-art Geomatic Engineering technology.
- Contributed to collaborative initiative with Oforikrom Municipal Assembly, aligning geospatial data with planning schemes to support Lands Commission objectives.
- Applied GNSS surveying techniques and community mapping approaches for urban planning and land administration.

Research Assistant
Regional Transport Research and Education Centre Kumasi (TRECK) — Kumasi, Ghana
Supervisor: Prof. Samuel Ato Andam-Akorful

- Presented research on **Smartphone-Based Road Condition Monitoring** at annual TRECK conference, awarded **2nd place** for presentation quality and technical contribution.
- Contributed to writing research reports and reviewing academic papers on transport and geospatial topics.
- Assisted students through tutorials and lab sessions while grading assignments and providing constructive feedback.

PROFESSIONAL EXPERIENCE

Remote Sensing, GIS & Software Developer
Afarinick Company Limited — Accra, Ghana

- Engineered backend systems integrating CNN-based deep learning models for real-time crop disease detection, gaining experience in deploying computer vision models to production environments.
- Designed PostgreSQL/PostGIS schemas storing farm geolocation, detection history, and disease classifications—learning how spatial data architecture supports machine learning pipelines.
- Built IoT data pipelines ingesting soil moisture and temperature readings, combining satellite-derived insights with ground sensor data for multi-modal agricultural monitoring.
- Implemented geospatial analysis for farm mapping and yield prediction, working with farm polygons, district boundaries, and route tracking in GeoDjango.

Computer Vision & Sensor Fusion Developer — Pothole Detection System
TRECK Innovation Challenge, Kumasi, Ghana

- Designed and developed a pothole detection mobile app integrating calibrated camera data with IMU sensors for physical dimension estimation.
- Integrated YOLOv8-based computer vision models for real-time pothole detection with automated logging of type, dimensions, and GPS coordinates.
- Combined multimedia capture with geolocation features and RESTful API integration for centralized data reporting.

Assistant Mine Surveyor (Intern)
Asanko Gold Ghana Ltd — Accra, Ghana

- Conducted high-precision topographic surveys using Total Stations and GPS to establish ground-control points for validating and calibrating satellite-derived geological models.
- Operated Terrestrial Laser Scanners (TLS) and drones to collect dense 3D point clouds, learning to interrogate high-resolution sensor data against observed geological structures.
- Processed and analyzed TLS, drone, and GPS data in QGIS/ArcGIS to generate mine maps and volumetric models, bridging field measurements with regional remote sensing datasets.

Geospatial Developer & GIS Analyst
Rainforest Alliance — Accra, Ghana

- Developed a hybrid monitoring system for Sui River cocoa-forest landscape covering 333.9 km^2 in Western North Region, Ghana.
- Implemented participatory landscape approach integrating community-based data collection with automated validation and error reduction mechanisms.
- Built data collection modules enabling local communities to contribute to forest monitoring while maintaining data quality through automated checks.

Geospatial Developer
Ghana EXIM Bank — Accra, Ghana

- Architected and maintained a distributed geospatial database using PostGIS with custom indexing and validation pipelines for high-integrity spatial data processing.
- Designed and built a RESTful Geoservices API implementing OGC standards to power web and mobile mapping applications with real-time spatial queries.
- Engineered scalable ETL frameworks and a validation engine automating ingestion and cleaning of diverse geodata sources, becoming the standard workflow for data integrity across field and analytics teams.

LEADERSHIP EXPERIENCE

Team Lead — Road Condition Monitoring AI Project
TRECK, Kumasi — Supervisor: Prof. Samuel Ato Andam-Akorful

- Led a team of three developing computer vision components for an AI-based road monitoring application.
- Built real-time pothole detection and dimension estimation systems using Python and computer vision.
- Recognized by supervisor for organizational skills, adaptability, and effective collaboration under pressure.

Judicial Committee Member
Ghana Institution of Surveyors Students Society (GhIS-SS)

- Enforce society constitution through fair disciplinary proceedings and dispute resolution.
- Contribute to policy development aligned with professional surveying ethics.
- Facilitate communication between members and leadership to address concerns.

PROJECTS

Afarisense – AI-Powered Cocoa Disease Detection Platform
Afarinick Company Limited

- Built a full-stack Django application integrating CNN-based deep learning models for real-time detection of Cocoa Swollen Shoot Virus Disease (CSSVD) with 92.44% accuracy.
- Discovered that models trained on commercial farms failed catastrophically on smallholder farms, learning shortcuts like uniform canopy structure rather than actual disease signatures, directly encountering algorithmic bias in geospatial ML.
- Addressed shortcut learning by diversifying training data and implementing augmentation strategies to force the model to focus on disease-specific features rather than environmental confounders.
- Designed PostgreSQL/PostGIS schema storing detection history, disease classifications, and farm geolocation for spatial analysis of outbreak patterns.

Phone Camera Calibration for Geospatial Mapping
Research Project — Supervisor: Prof. Samuel Ato Andam-Akorful

- Developed calibration algorithms to correct lens distortions in smartphone cameras, establishing how sensor characteristics affect geospatial data quality before machine learning analysis.
- Investigated how raw sensor outputs deviate from physical reality, gaining insight into measurement errors that propagate through computer vision pipelines.
- Created reproducible workflows for distortion correction, emphasizing data provenance as foundational for robust machine learning.

ACL – Intelligent Farm Management System
Afarinick Company Limited

- Architected modular enterprise platform unifying GIS, IoT telemetry, and satellite-derived insights for smallholder farm monitoring.
- Engineered IoT pipeline ingesting soil moisture and temperature readings, combining ground sensor data with satellite imagery for multi-modal agricultural monitoring.
- Designed complex GeoDjango/PostGIS models with farm polygons (MultiPolygonField), district boundaries, and route tracking, learning the importance of spatial data architecture for geospatial ML applications.
- Experienced firsthand the labor-intensive work of acquiring and preprocessing satellite data, reinforcing belief in the need for shared benchmark datasets like Fields of the World.

Pothole Detection System — Multi-Sensor Fusion for Physical Measurement
Undergraduate Thesis — Supervisor: Prof. Samuel Ato Andam-Akorful

- Developed smartphone-based road condition monitoring system combining calibrated camera data with IMU sensors to estimate physical pothole dimensions using YOLOv8.
- Observed significant model degradation when moving from highways to unpaved rural roads—an early encounter with model generalization challenges across diverse environments.
- Noted vision model relied on visual shortcuts (paved texture, lane markings) while IMU data provided physically-grounded measurements consistent across environments.
- Recognized that integrating physical sensor data with vision models offers a pathway to more robust predictions, directly informing interest in physically-informed geospatial machine learning.

Sui River Cocoa-Forest Landscape Monitoring System
Rainforest Alliance

- Developed a hybrid monitoring system for 333.9 km^2 cocoa-forest landscape in Western North Region, Ghana, integrating mobile data collection with real-time GIS dashboards.
- Implemented participatory tools empowering local communities to contribute to forest monitoring while maintaining data quality through automated validation.
- Tracked deforestation, cocoa cultivation, and livelihood indicators using spatial data, creating evidence base for conservation decisions.

E-Digital GIS Solution for GEXIM Funded Farms
Ghana Export-Import Bank

- Architected a comprehensive GIS platform for tracking development projects and loan portfolios across agricultural landscapes.
- Built a mobile data collection application with offline-capable field monitoring, verification, and GIS mapping capabilities.
- Implemented project performance indicators with automated data validation and real-time dashboards for monitoring outcomes.
- Created a beneficiary tracking system integrating spatial data with impact assessment for development effectiveness.

Mine Surveying — Ground Truth for Satellite Models
Asanko Gold Ghana Ltd (Internship)

- Conducted high-precision topographic surveys using Total Stations and GPS to establish ground-control points for validating satellite-derived geological models.
- Operated Terrestrial Laser Scanners (TLS) and drones to collect dense 3D point clouds, learning to interrogate high-resolution sensor data against observed geological structures.
- Processed and analyzed TLS, drone, and GPS data in QGIS/ArcGIS, bridging precise field measurements with regional remote sensing datasets.
- Gained understanding of how ground truth data validates and calibrates satellite-based models—essential knowledge for robust remote sensing ML.

Geospatial Summer Camp — Ayigya Town Mapping Project
KNUST — Lead: Prof. Samuel Ato Andam-Akorful

- Participated in an intensive two-week field camp conducting hands-on data collection of natural and man-made features using GNSS surveying technology.
- Contributed to collaborative initiative with Oforikrom Municipal Assembly, aligning geospatial data with town planning schemes to support Lands Commission objectives.
- Applied community mapping approaches for urban planning and land administration in real-world governance context.

TECHNICAL SKILLS

- **Geospatial & Remote Sensing**: QGIS, ArcGIS Pro, Google Earth Engine, PostGIS, Drone2Map, Agisoft MetaShape
- **Machine Learning & Computer Vision**: TensorFlow, PyTorch, Scikit-Learn, OpenCV
- **Programming Languages**: Python, R, JavaScript (for Earth Engine), SQL
- **Data Science & Analytics**: Pandas, NumPy, Jupyter, Google Colab, R (tidyverse)
- **Surveying & Field Data Collection**: Total Stations, GPS/GNSS, Terrestrial Laser Scanning (TLS), Drone Operations
- **Database Systems**: PostgreSQL/PostGIS
- **Development Tools**: Git, Linux/Unix
- **Documentation**: LaTeX

HONORS AND AWARDS

- **Best Final Year Project Honors – Highest Departmental Score (88%)**
  Department of Geomatic Engineering, Kwame Nkrumah University of Science and Technology
  Received top honors for highest-scoring final year project: "Phone Camera Calibration for Mapping Applications," a core component of research on smartphone-based road condition monitoring. Supervised by Prof. Samuel Ato Andam-Akorful.
- **First Class Honors in Geomatic Engineering**
  Kwame Nkrumah University of Science and Technology
  Graduated with First Class Honors, recognized among top-performing students for outstanding academic achievement.
- **Ghana Institution of Surveyors Students Society (GhIS-SS) Award**
  Recognized for academic excellence and contributions to the surveying profession through service as Judicial Committee Member and Software Engineer.
- **Second Place – TRECK Conference Presentation**
  Regional Transport Research and Education Centre Kumasi
  Awarded second place for research presentation quality and technical contribution.

REFERENCES

- **Prof. Samuel Ato Andam-Akorful**
  Associate Professor, Department of Geomatic Engineering
  Kwame Nkrumah University of Science and Technology (KNUST), Kumasi, Ghana
  Email: saandam-akorful.coe@knust.edu.gh
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